

Size composition and exploitation pattern of Gangetic Tengra from the tributary of the Ganga river at Ayodhya, Uttar Pradesh, India

Shakila Khan¹, Awanindra Kumar Tiwari², Priyanka Mayank³ and Sarita Tripathi³

¹College of Fisheries, Narendra Dev University of Agriculture and Technology, Kumarganj, Ayodhya, Uttar Pradesh, India

²Krishi Vigyan Kendra, Raebareli, Uttar Pradesh, India

³Kalash Research and Welfare Society, Prayagraj, Uttar Pradesh, India

Accepted : 16 February 2019

Abstract

Fishing pressure changes the size composition, growth rate, age composition, sex ratio, biodiversity, income of fisher/fishermen and maturation of fishes. The estimation of size composition and exploitation pattern of Gangetic Tengra, *Sperata aor* total length varied from 9.1 to 119.0 cm in November 2017 to October 2018 from the Sarayu river at Ayodhya, Uttar Pradesh, India. The minimum exploitation was estimated in 109.1 to 119.0 cm size group with 1.14%) while maximum exploitation was estimated in 59.1 to 69.0 cm size group (19.70%). The exploitation was observed 3.41% in 9.1-19.0 cm, 6.44% in 19.1-29.0 cm, 10.61% in 29.1-39.0 cm, 13.26% in 39.1-49.0 cm, 17.80% in 49.1-59.0 cm, 14.39% in 69.0-79.1 cm, 4.54% in 89.0-99.0 cm and 2.65% in 99.0-109.0 cm size groups. Results indicated that the exploitation was not systematic in middle size group to higher size group. Overall, the middle size group contributed maximum exploitation compared to lower and higher size group.

Keywords: Sarayu river, *Sperata aor*, Size composition, Gangetic Tengra, Exploitation pattern

Introduction

The size composition of fishes is an essential component of river and stream ecosystem and represents an environmental structure, function, depth, food supply, food web and health of streams/ rivers and possibility of recruitments (Nautiyal *et al.* 2004, 2007, Dwivedi *et al.* 2008, 2023, Imran *et al.* 2015, Tiwari *et al.* 2016, 2020, Mishra & Dwivedi 2022, Dwivedi & Mishra 2023a). *Sperata aor* (Gangetic Tengra) is a very common fish species from the Ganga river and its tributaries (Dwivedi *et al.* 2016a, Tripathi *et al.* 2017a, Nautiyal & Dwivedi 2019, Prakash *et al.* 2019). They prefer riverine ecosystem (Chondar 1999, Dwivedi *et al.* 2019). The riverine ecosystem is the base of biodiversity especially fishes and aquatic insects in fresh water habitat (Nautiyal *et al.* 2004, Mishra & Dwivedi 2022). It constitutes the chief component of the fishery in the middle and lower stretches of the river (Talwar & Jhingran 1991, Tripathi *et al.* 2017b) and its tributaries (Mayank & Dwivedi 2015a, Nautiyal & Dwivedi 2019). It is a very popular food fish in the Ganga basin due to its good taste and high nutritional value (Mayank *et al.* 2018). Wild stock of *S. aor* is more abundant compared to *S. seenghala* in the nature/wild ecosystem

(Example rivers and streams). Its natural distribution is in the freshwater ecosystem of India, Pakistan, Nepal, Bangladesh, Sri Lanka and Burma (Chondar 1999).

The objective of the present study was to give size composition and exploitation of the *S. aor* from the Sarayu river at Ayodhya, Uttar Pradesh, India.

Material and methods

Random fish samples of *Sperata aor* were collected between November 2017 to October 2018 from the Sarayu river at Ayodhya, Uttar Pradesh, India. It is a major left bank tributary of the Ganga river. Fishes were captured using a variety of methods including gill netting, drag netting (mahajal, chaundi, darwari), cast netting and hook and line. Size of fishes was measured by simple measuring scale. The total length (TL) of fish was measured from tip of the snout to the largest rays of caudal fin. Collected data were classified at 10 cm intervals and arranged a systematic series. The number of samples calculated according to size group then converted into percentage for estimation of exploitation pattern.

Result and discussion

The size composition of *Sperata aor* (Gangetic Tengra) varied from 9.1 to 119.0 cm

from the Sarayu river at Ayodhya, Uttar Pradesh, India (Table 1). The highest exploitation was observed in 59.1 to 69.0 cm size group (19.70%) while lowest exploitation was recorded in 109.1-119.0 cm size group with 1.14%. The size groups 9.1-19.0 cm, 19.1-29.0 cm, 29.1-39.0 cm, 39.1-49.0 cm, 49.1-59.0 cm, 69.1-79.0 cm, 79.1-89.0 cm, 89.1-99.0 cm and 99.1-109.0 cm were recorded exploitation of 3.41%, 6.44%, 10.61%, 13.26%, 17.80%, 14.39%, 6.06%, 4.54% and 2.65%, respectively (Table 1).

Overall, middle size group contributed maximum exploitation compared to lower size group and higher size group. Results also indicated that the exploitation was not systematic in middle size group to higher size group. If higher size group exploitation is not systematic then it is an alarming sign for future population of *S. aor* in declining order. In collective, middle size group of fishes in case of riverine/natural stocks are more exploited than other size groups, it is harmful for the stock of fishes (Nikolskii 1980, Pathak *et al.* 2011, 2015, Tripathi *et al.* 2015).

The middle size groups of any commercially important fish species have more attractive for the consumers and fishers (Imran

et al. 2015, Mayank & Dwivedi 2015b, 2016a, Dwivedi *et al.* 2016b). Similar findings were recorded by Tiwari *et al.* (2016, 2017) and Dwivedi *et al.* (2018) from the Ganga river. The catfish fishery is the backbone from the middle stretch of the Ganga river in current scenario (Mayank & Dwivedi 2015b, Dwivedi *et al.* 2016a). The water discharge of the river also interferes in the production of the culture pond of the surrounding river bank (Jha *et al.* 2015, Dwivedi *et al.* 2018). Exploitation is an economic activity governed by social needs and pressures (Dwivedi *et al.* 2014, Mayank & Dwivedi 2016b, Gopesh *et al.* 2021). In case of economically important fishes, over exploitation, water pollution and targeted fishing is the main problem for declining of the riverine fishery stock (Jatan *et al.* 2014, Tiwari *et al.* 2016, Dwivedi *et al.* 2016c, 2017a, Muthmainnah & Gaffar 2017, Mayank *et al.* 2018, Mishra and Dwivedi 2020, 2022, Mishra *et al.* 2023b). The fishing pressure, size of nets, mesh size and fishing technique (example degree) are responsible for increasing or decreasing of the total length of fishes and recruitment pattern in the lotic ecosystems (Dwivedi *et al.* 2016b, 2017b, Dwivedi & Mishra 2023).

References

- Chondar, S.L. (1999). Biology of Finfish and Shellfish. India: SCSC Publishers. pp. 514.
- Dwivedi, A. C., P. Mayank & R. K. Pathak (2016b.) Size composition and exploitation structure of Indian major carp, *Cirrhinus mrigala* (Hamilton, 1822) from the Ganga river, India. *Journal of Fisheries and Life Science*, **1(1)**: 30-32.
- Dwivedi, A. C. & N. Mishra (2023). Health of stock of commercially exploited non native fish species, *Cyprinus carpio* (Linnaeus, 1758) from the tributary of the Ganga River, Central India. *Geology, Earth and Marine Sciences*, **5(1)**: 1-3, DOI: 10.31038/GEMS.2023514.
- Dwivedi, A.C. & N. Mishra (2023). Health of stock of commercially exploited non native fish species, *Cyprinus carpio* (Linnaeus, 1758) from the tributary of the Ganga River, Central India. *Geology, Earth and Marine Sciences*, **5(1)**: 1-3, DOI: 10.31038/GEMS.2023514.
- Dwivedi, A.C., A. S. Mishra, P. Mayank & A. Tiwari (2016a). Persistence and structure of the fish assemblage from the Ganga river (Kanpur to Varanasi section), India. *Journal of Geography and Natural Disasters*, **6**: 159. Doi: 10.4172/2167- 0587.1000159.
- Dwivedi, A.C., A. Tiwari & P. Mayank (2018). Environmental pollution supports to constancy and invader potential of *Cyprinus carpio* and *Oreochromis niloticus* from the Ganga river, India. *International Journal of Poultry and Fisheries Sciences*, **2(1)**: 1-7.
- Dwivedi, A.C., A. S. Mishra, S. Prakash, B. K. Mishra, P. Mayank, A. Tiwari & J. Kumar (2019). Current status of Gangetic tengra from the middle stretch of the Ganga river, India. *Journal of Fisheries and Life Science*, **4(1)**: 5-8.

- Dwivedi, A.C., D. N. Jha & P. Mayank (2014). Food security, livelihood and non-native fish species: status, trends and future Perspectives. *Journal of the Kalash Science*, **2(3): (Special Volume)**: 41-46.
- Dwivedi, A.C., K. R. Singh, S. Khan & P. Mayank (2008). Dynamics of exploited fish populations and sex ratio of *Cyprinus carpio* var. *communis* (Linnaeus) in the Yamuna river at Allahabad. *The Asian Journal of Animal Sciences*, **3(2)**: 198-202.
- Dwivedi, A.C., N. Mishra & P. Mayank (2023). Variations in the catchability and remarkable size-spectra stability of *Cyprinus carpio* (Linnaeus, 1758) with reference of ecology from the Tons river, Central India. *Ecology and Conservation Science: Open Access*, **2(3)**: 555590. DOI: 10.19080/ECO.A.2023.02.555590.
- Dwivedi, A.C., P. Mayank & A. Tiwari (2016c). The River as transformed by human activities: the rise of the invader potential of *Cyprinus carpio* and *Oreochromis niloticus* from the Yamuna River, India. *Journal of Earth Science & Climatic Change*, **7(7)**: 361. doi: 10.4172/2157-7617.1000361.
- Dwivedi, A.C., P. Mayank & A. Tiwari (2017b). Size selectivity of active fishing gear: changes in size, age and growth of *Cirrhinus mrigala* from the Ganga River, India. *Fisheries and Aquaculture Journal*, **8(2)**: DOI: 10.4172/2150-3508.1000205.
- Dwivedi, A.C., P. Mayank, S. Tripathi & A. Tiwari (2017a). Biodiversity: the non-natives species versus the natives species and ecosystem functioning. *Journal of Biodiversity, Bioprospecting and Development*, **4(1)**: DOI: 10.4172/2376- 0214.1000164.
- Gopesh, A., S. Tripathi, K. D. Joshi & A. C. Dwivedi (2021). Size composition, exploitation structure and sex ratio of *Clupisoma garua* (Hamilton) from middle stretch of the Ganga River at Allahabad, India. *National Academy Science Letter*, **44(4)**: 309-311, DOI: 10.1007/s40009-020-01011-0.
- Imran, S., Sasya Thakur, D. N. Jha & A. C. Dwivedi (2015). Size composition and exploitation pattern of *Labeo calbasu* (Hamilton 1822) from the lower stretch of the Yamuna river. *Asian Journal of Bio Science*, **10(2)**: 162-164. DOI: 10.15740/HAS/AJBS/10.2/171-173.
- Jatan, I., H.O. Barot, & I.A. Fahabi (2014). Cross-examination on taxonomic status, economic importance, geographical distribution, present status and prevailing threats of fish varieties, Himalayan and Indo Burma biodiversity hotspots zones of North East India, *Afr. J. Fish. Sci.*, **2(6)**: 107-127.
- Jha, D.N., K. D. Joshi, A. C. Dwivedi, P. Mayank, M. Kumar & A. Tiwari (2015). Assessment of fish production potential of Chitrakoot district, Uttar Pradesh. *Journal of the Kalash Science*, **3(3, Special Volume)**: 7-10.
- Mayank, P. & A. C. Dwivedi (2016a). Stock assessment and population structure of alien fish species, *Oreochromis niloticus* (Linnaeus) from the lower stretch of the Yamuna river, India. *Journal of the Experimental Zoology, India*, **19(1)**: 163- 167.
- Mayank, P. & A. C. Dwivedi (2016b). Linking *Cirrhinus mrigala* (Hamilton, 1822) size composition and exploitation structure to their restoration in the Yamuna river, India. *Asian Journal of Bio Science*, **11(2)**: 292-297. doi: 10.15740/HAS/AJBS/11.2/292-297.
- Mayank, P. & A. C. Dwivedi (2015b). River health and commercially important catfishes from the Yamuna river, India. *Journal of the Kalash Science*, **3(3, Special Volume)**: 23-26.
- Mayank, P. & A.C. Dwivedi (2015a). Biology of *Cirrhinus mrigala* and *Oreochromis niloticus*. LAP LAMBERT Academic Publishing GmbH & Co. KG, Dudweiler Landstr. 99, 66123 Saarbrücken, Germany, Pp. 188.
- Mayank, P., A. C. Dwivedi & R. K. Pathak (2018). Age, growth and age pyramid of exotic fish species *Oreochromis niloticus* (Linnaeus 1758) from the lower stretch of the Yamuna river, India. *National Academy Science Letter*, **41(6)**: 345-348. doi.org/10.1007/s-40009-018-0673-7.
- Mishra N. & A. C. Dwivedi (2022). Reproductive properties and impacts of invasive alien species, Common carp (*Cyprinus carpio* Linnaeus, 1758) from the Tons river at Prayagraj, India. *Journal of the Kalash Science*, **10(2)**: 5-11.
- Mishra N. & A. C. Dwivedi (2020). Environmental drivers supports to distribution, composition and biology of *Cyprinus carpio* (Linnaeus, 1758) in respect of time scale: A review. *Journal of the Kalash Science*, **8(2)**: 91-102.
- Mishra N., A. C. Dwivedi & P. Mayank (2023). Studies on age profile of Common carp, *Cyprinus carpio* (Linnaeus, 1758) from the Tons River, India. *Journal of Marine Science and Research*, **2(1)**. DOI: 10.58489/28365933/006.

- Mishra, A.S. & A. C. Dwivedi (2022). Taxonomic and trophic characterization of benthic macroinvertebrate fauna in headwaters of eastern Himalaya. *Indian Journal of Ecology*, **49(4)**: 1497-1502 DOI: 10.55362/IJE/2022/3689.
- Muthmainnah, D. & A. K. Gaffar (2017). Fish and fisheries in flood plain swamp in middle part of Musi river. *Indonesian J. Environ. Manag. Sustainability*, **1(1)**: 1-5.
- Nautiyal, P. & A. C. Dwivedi (2019). Fishery in the tributaries of Yamuna river (Ken river, Paisuni rivers) and Ganga river (Tons river). *J. Mountain Res.*, **14(2)**: 19-36.
- Nautiyal, P., A. C. Dwivedi, A. Shivam & K. R. Singh (2007). Possibility of breeding grounds of Mahseer in the Paisuni R. (Chitrakoot Dham): its ecology, and status of *Tor tor* (Hamilton) in the north Vindhyan rivers. *J. Bombay Natural Hist. Soc.*, **104(3)**: 355-357.
- Nautiyal, P., A. Shivam, G. Rawat, Krishna Raj Singh, J. Varma & A. C. Dwivedi (2004). Longitudinal variation in the structure of benthic communities in the upland Vindhyan and Himalayan Rivers: River Continuum Concept approach. *National Journal of Life Sciences*, **1(1)**: 85-88.
- Nikolskii, G.V. (1980). Theory of fish population dynamics as the biological background for rational exploitation and management of fishery resources. Bishen Singh Mahendra Pal Singh Dehra Dun (India) and Otto Koeltz Science Publishers Koenigstein, W. Germany, pp. 1-323.
- Pathak, R. K., A. Gopesh & A. C. Dwivedi (2011). Alien fish species, *Cyprinus carpio* var. *communis* (common carp) as a powerful invader in the Yamuna river at Allahabad, India. *National Academy of Science Letter*, **34(9 & 10)**: 367-373.
- Pathak, R.K., A. Gopesh & A.C. Dwivedi (2015). Invasion potential and biology of *Cyprinus carpio* (Common carp) LAP LAMBERT Academic Publishing GmbH & Co. KG, DudweilerLandstr.99, 66123 Saarbrücken. Germany.
- Prakash, S., P. Mayank, A. C. Dwivedi, B. K. Mishra, A. Tiwari & S. K. Upadhyay (2019). Size structure and exploitation pattern of *Rita rita* (Hamilton 1822) in the Yamuna river at Prayagraj, Uttar Pradesh. *Journal of the Kalash Science*, **7(2)**: 21-24.
- Talwar, P.K. & A.G. Jhingran (1991). Inland Fishes of India and Adjacent Countries. 1st Ed. New Delhi, Bombay and Calcutta: Oxford and IBH Publishing Co. Pvt. Ltd. Pp. 1063.
- Tiwari, A., A. C. Dwivedi & P. Mayank (2016). Time scale changes in the water quality of the Ganga River, India and estimation of suitability for exotic and hardy fishes. *Hydrology Current Research*, **7(3)**: 254. doi:10.4172/2157-7587.1000254.
- Tiwari, A., A. C. Dwivedi & M. A. Rahman (2020). Assessment of heavy metal accumulation in vital tissues of commercially exploited fish, *Cyprinus carpio* (common carp) from the river Ganga, India. *Journal of Emerging Technologies and Innovative Research*, **7(6)**: 959-972.
- Tiwari, A., P. Mayank & A. C. Dwivedi (2017). Assessment of human health risk via the consumption of the freshwater fish, *Cyprinus carpio* and *Oreochromis niloticus* from the Ganga River, India. *Bioved*, **28(2)**: 341-349.
- Tripathi, S., A. Gopesh & A. C. Dwivedi (2017b). Framework and sustainable audit for the assessing of the Ganga river ecosystem health at Allahabad, India. *Asian Journal of Environmental Science*, **12(1)**: 37-42. DOI: 10.15740/HAS/AJES/12.1/37-42.
- Tripathi, S., A. Gopesh & A. C. Dwivedi (2017a). Fish and fisheries in the Ganga river: current assessment of the fish community, threats and restoration. *Journal of Experimental Zoology, India*, **20(2)**: 907-912.
- Tripathi, S., A. Gopesh, K.D. Joshi & A. C. Dwivedi (2015). Size composition, exploitation pattern, sex ratio and sex structure of *Eutropiichthys vacha* (Hamilton, 1822) from the middle stretch of the river Ganga at Allahabad, India. In: *Advances in biosciences and Technology* edited by K.B. Pandeya, A.S. Mishra R.P. Ojha and A.K. Singh published by NGBU, Allahabad, Pp 116-120.

Table 1 Size composition and exploitation pattern of *Sperata aor* (Gangetic Tengra) from the Sarayu river at Ayodhya, Uttar Pradesh

SN	Size groups (cm)	No. of samples	Percentage
1	9.1-19.0	09	3.41
2	19.1-29.0	17	6.44
3	29.1-39.0	28	10.61
4	39.1-49.0	35	13.26
5	49.1-59.0	47	17.80
6	59.1-69.0	52	19.70
7	69.1-79.0	38	14.39
8	79.1-89.0	16	6.06
9	89.1-99.0	12	4.54
10	99.1-109	07	2.65
11	109.1-119.0	03	1.14